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MMRM

 Will be reviewed

8

A NAT translation table typically maps:

- ☐ Public IP address --> Private IP address only
- ☐ Public port number --> Private port number only
- ☒ (Private IP address, Private port number) <--> (Public IP address, Public port number)
- ☐ Private IP address --> Public IP address only
- ☐ Private port number --> Public port number only
- ☐ None of the above

 Will be reviewed

9

Suppose that there is exactly one packet switch "X" between a sending host "A" and a receiving host "B". The transmission rate of the link between "A" and "X" as well as the link between "X" and "B" is R bps. Assume that the packet switch "X" uses cut-through. If the processing, queuing, and propagation delays are ignored, what is the total end-to-end delay to send a packet of length L bits from "A" to "B"?

- ☐ Zero
- ☐ $2L/R$
- ☐ $2R/L$

- ☒ L/R
- ☐ R/L
- ☐ None of the above

 **Will be reviewed**

10

Which of the following statements correctly describe the advantages and disadvantages of switches and routers?

- ☐ Switches always require manual configuration and cannot operate at full-duplex. Routers eliminate collisions and are faster because they process packets only up to link layer.
- ☐ Switches support hierarchical addressing and firewall protection. Routers are plug-and-play and eliminate collision.
- ☒ Switches are plug-and-play, do not offer any protection against broadcast storms, and the active topology of a switched network is restricted to spanning tree. Routers have a larger per-packet processing time than switches, support hierarchical addressing, but require configuration.
- ☐ There is no restriction on switched network topology. But the active topology of the network that uses routers is restricted to spanning tree.
- ☐ None of the above

 **Will be reviewed**

11

Assume Host A sends five consecutive data segments to Host B using GBN, or TCP (no delayed ACK). The timeout values for all protocols are sufficiently long such that all five data segments and their corresponding ACKs can be received by both hosts before any timeout event, if not lost in the channel. Suppose the second ACK from host B is lost and no further losses or errors occur after that. In the end, all five data segments have been correctly received by Host B. Which of the following correctly describes the total segments sent by Host A and total ACKs sent by Host B?

- ☐ Number of (GBN, TCP) Segments sent by A is (GBN→6, TCP→6); Number of ACKs sent by B is (GBN→5, TCP→5)

- ☐ Number of (GBN, TCP) Segments sent by A is (GBN→9, TCP→6); Number of ACKs sent by B is (GBN→7, TCP→5)
- ☐ Number of (GBN, TCP) Segments sent by A is (GBN→5, TCP→5); Number of ACKs sent by B is (GBN→6, TCP→6)
- ☐ Number of (GBN, TCP) Segments sent by A is (GBN→9, TCP→6); Number of ACKs sent by B is (GBN→5, TCP→5)
- ☐ Number of (GBN, TCP) Segments sent by A is (GBN→5, TCP→5); Number of ACKs sent by B is (GBN→5, TCP→5)
- ☐ Number of (GBN, TCP) Segments sent by A is (GBN→9, TCP→6); Number of ACKs sent by B is (GBN→8, TCP→5)
- ☒ Number of (GBN, TCP) Segments sent by A is (GBN→9, TCP→5); Number of ACKs sent by B is (GBN→8, TCP→5)
- ☐ None of the above

 **Will be reviewed**

12

Which of the following correctly describes the characteristics of switching fabrics inside a router?

- ☐ Switching via memory and bus forwards multiple packets simultaneously.
- ☐ Switching via bus is faster than switching via interconnection network.
- ☒ Switching via bus forwards only one packet at a time, while switching via standard interconnection network can forward multiple packets in parallel unless they target the same output.
- ☐ Switching via memory forward packets without routing processor intervention.
- ☐ All of the above
- ☐ None of the above

 **Will be reviewed**

13

Which of the following best distinguishes forwarding from routing in the internet network layer?

- ☒ Forwarding moves packets within a router; routing determines best end-to-end paths.
- ☐ Forwarding is network-wide; routing is router-local.
- ☐ Forwarding determines paths; routing moves packets between nodes on these paths.
- ☐ Forwarding is fixed; routing is dynamic.
- ☐ None of the above

 **Will be reviewed**

14

Why does a DHCP client initially send its discover message to 255.255.255.255?

- ☐ Because the client does not yet know the IP address of DHCP server.
- ☒ Because the DHCP server only listens to broadcast messages with broadcast IPs.
- ☐ Because DHCP server does not know the IP address of the client.
- ☐ None of the above

 **Will be reviewed**

15

Which of the following statements correctly describes circuit switching?

- ☐ Circuit switching allows multiple users to share the same channel dynamically, providing high efficiency during idle periods, but it cannot guarantee bandwidth or delay.
- ☐ Circuit switching eliminates the need for a dedicated path, uses packet-based transmission, and is highly efficient, but it suffers from high delay during congestion.

☐ Circuit switching provides no dedicated resources, allows dynamic bandwidth allocation, and guarantees low delay, but is prone to packet loss.

☒ Circuit switching can provide a dedicated path with guaranteed bandwidth and low delay for the duration of the connection, but it can be inefficient if the circuit is idle, and call setup introduces additional delay.

☐ None of the above

 **Will be reviewed**

16

Which of the following is a disadvantage of TDM and FDM?

☐ Collisions are frequent and unavoidable.

☐ Nodes must use random backoff algorithm.

☐ A master node is needed.

☒ A node is limited to use a fraction of the channel even if it is the only active node.

☐ None of the above

 **Will be reviewed**

17

Which of the following correctly lists Cookie technology components in the Web?

☐ A cookie header in the HTTP response; a cookie header in the HTTP request; a proxy cache that stores cookies; and a DNS database at the server.

☐ A cookie file in the HTTP response message; a cookie file in the HTTP request message; a cookie file stored on the user's end system; and a back-end database at the Web site.

☐ A cookie header line in the HTTP response message; a cookie header line in the HTTP request message; a shared proxy cache managed by the ISP; and a back-end database at the Web site.

- ☒ A cookie header line in the HTTP response; a cookie header line in the HTTP request; a cookie file kept on the user's end system and managed by the user's browser; and a back-end database at the Web site.
- ☐ A cookie header line in the HTTP response; a cookie header line in the HTTP request; a cookie file kept on the user's end system; and a back-end DNS database at the server.
- ☐ None of the above

📝 **Will be reviewed**

18

Which of the following correctly describes DNS servers?

- ☐ Root servers store all DNS records; TLD servers map hostnames; authoritative servers are connected directly to local servers, which are connected to local users.
- ☒ Root servers provide the IP addresses of TLD servers; TLD servers provide the IP addresses of authoritative servers; authoritative servers maintain host-to-IP mappings; and local servers are connected to local users.
- ☐ Root servers map hostnames for TLD servers; TLD servers map hostnames for authoritative servers; and authoritative servers provide IP addresses of local servers.
- ☐ Root servers maintain host-to-IP mappings of TLD servers; TLD servers maintain host-to-IP mappings of authoritative servers; authoritative servers maintain host-to-IP mappings of local servers; and local servers maintain host-to-IP mappings of local users.
- ☐ None of the above

📝 **Will be reviewed**

19

A client retrieves a web page consisting of one base HTML file and two embedded objects, all stored on the same web server. The size of the HTML file and each embedded object is 100 Kbits. The round-trip time between the client and the server is 50 ms, and the data transmission rate is 10 Mbps. The client uses persistent HTTP with pipelining. Assume that the transmission time of control packets (TCP three-way handshake, HTTP requests, and acknowledgments) is negligible. What is the minimum time required for the client to completely receive the entire web page?

- ☐ 80 ms
- ☐ 130 ms
- ☐ 150 ms
- ☒ 180 ms
- ☐ 200 ms
- ☐ 230 ms
- ☐ 330 ms
- ☐ None of the above

 **Will be reviewed**

20

In BitTorrent, why does a peer request the rarest chunk first from its neighbors?

- ☐ To maximize the peer's individual download speed by selecting chunks that arrive fastest.
- ☐ To prioritize sending chunks to peers that upload the fastest to others.
- ☐ To give priority to chunks uploaded directly from the original server.
- ☐ To force peers to remain in the torrent until everyone finishes.
- ☒ To ensure that the rarest chunks are more quickly redistributed and are obtained early before the peers that own them leave the torrent.
- ☐ None of the above

 **Will be reviewed**

21

In TCP, if the receiver window is zero, what can the sender do?

- ☒ Stop sending segments until the receiver advertises a nonzero window
- ☐ Continue to send segments with one data byte until the receiver advertises a nonzero window
- ☐ Continue sending normal segment size
- ☐ Ask the receiver to increase its window size
- ☐ None of the above

 **Will be reviewed**

22

A host "X" on Subnet 1 wants to send a datagram to a host "Y" on Subnet 2. Which MAC address will the ARP module in host "X" request if it does not already have it?

- ☐ The MAC address of the host "Y" interface
- ☒ The MAC address of the last-hop router interface on Subnet 2
- ☐ The MAC address of the first-hop router interface on Subnet 1
- ☐ None of the above

 **Will be reviewed**

23

Which of the following statements about FTP is correct?

- ☐ FTP uses a single TCP connection for both control and data.
- ☒ The FTP control connection is persistent, but a new data connection is created for each file.
- ☐ FTP data connection remains open throughout the session.
- ☐ Both FTP control connection and data connection remain open throughout the session.

☐ None of the above

 **Will be reviewed**

24

Which of the following best explains HTTP and SMTP?

- ☒ HTTP connection can be persistent or non-persistent ; SMTP connection is persistent.
- ☐ SMTP sends each object in a separate response message; HTTP sends all objects in one message.
- ☐ SMTP and HTTP require 7-bit ASCII encoding.
- ☐ HTTP is push protocol; SMTP is pull protocol.
- ☐ All of the above
- ☐ None of the above

 **Will be reviewed**

25

Consider that a switch has the following forwarding table: MAC address 62-FE-F7-11-89-A3 is associated with interface 1, and MAC address 7C-BA-B2-B4-91-10 is associated with interface 3. A frame with source MAC address AA-BB-CC-DD-EE-FF arrives at interface 2 and has destination MAC address 11-22-33-44-55-66. What action does the switch take?

- ☐ The switch broadcasts the frame to all interfaces except interface 2 and adds a new entry to the forwarding table that associates the MAC address AA-BB-CC-DD-EE-FF with interface 2.
- ☐ The switch drops the frame (Filtering).
- ☐ The switch forwards the frame only to interface 2.
- ☒ The switch broadcasts the frame to all interfaces except interface 2 and adds a new entry to the forwarding table that associates the MAC address 11-22-33-44-55-66 with interface 2.

- ☐ The switch forwards the frame only to interface 2 and adds a new entry to the forwarding table that associates the MAC address 11-22-33-44-55-66 with interface 2.
- ☐ None of the above

 **Will be reviewed**

26

Question

- ☒ a)
- ☐ b)
- ☐ c)
- ☐ d)
- ☐ e)
- ☐ f)
- ☐ g)

 **Will be reviewed**

27

Which of the following is a key difference between TCP and UDP from the transport layer services perspective?

- ☐ TCP provides de-multiplexing; UDP does not.
- ☒ TCP provides reliable, connection-oriented service; UDP provides unreliable, connectionless service.
- ☐ UDP guarantees timing and throughput; TCP does not.
- ☐ UDP guarantees timing and throughput; TCP guarantees reliability and security.

☐ None of the above

 **Will be reviewed**

28

A router interconnects three subnets (A, B, and C), all using addresses from 192.168.10.0/24, where Subnet A requires at least 100 interfaces, Subnet B requires at least 50 interfaces, and Subnet C requires at least 20 interfaces. Which of the following sets of subnet network addresses satisfies all these constraints if oversized subnets are allowed?

- ☐ Subnet A uses 192.168.10.0/26, Subnet B uses 192.168.10.64/25, and Subnet C uses 192.168.10.192/28.
- ☐ Subnet A uses 192.168.10.0/25, Subnet B uses 192.168.10.128/26, and Subnet C uses 192.168.10.192/26.
- ☒ Subnet A uses 192.168.10.0/25, Subnet B uses 192.168.10.64/26, and Subnet C uses 192.168.10.128/27.
- ☐ Subnet A uses 192.168.10.0/26, Subnet B uses 192.168.10.64/26, and Subnet C uses 192.168.10.128/27.
- ☐ None of the above

 **Will be reviewed**

29

Question

- ☒ a)
- ☐ b)
- ☐ c)
- ☐ d)
- ☐ e)

☐ f)

 Will be reviewed

30

Question

☐ a)

☐ b)

☐ c)

☒ d)

☐ e)

☐ f)

 Will be reviewed

31

Consider the GBN protocol with a sender window size of 4 and a sequence number range of 1,024. Suppose that at time t , the next in-order packet that the receiver is expecting has a sequence number of k . Assume that the medium does not reorder messages. Which of the following is possible set of sequence numbers inside the sender's window at time t ?

☐ $\{k, k+1, k+2, k+3\}$

☒ $\{k-1, k, k+1, k+2\}$

☐ $\{k-2, k-1, k, k+1\}$

☐ $\{k-3, k-2, k-1, k\}$

☐ $\{k-4, k-3, k-2, k-1\}$

- ☐ All of the above
- ☐ None of the above

 **Will be reviewed**

32

Question

- ☒ Yes
- ☐ No

 **Will be reviewed**

33

An IP datagram of 3000 bytes (no options header) is transmitted over a link with an MTU of 1500 bytes. Which of the following option is correct?

- ☐ Number of Fragments = 3 ; Frag#1: Offset=0, MF=1 ; Frag#2: Offset=1500, MF=1 ;Frag#3: Offset=3000, MF=0
- ☐ Number of Fragments = 2 ; Frag#1: Offset=0, MF=1 ; Frag#2: Offset=185, MF=0
- ☐ Number of Fragments = 3 ; Frag#1: Offset=0, MF=0 ; Frag#2: Offset=1480, MF=0 ; Frag#3: Offset=2960, MF=1
- ☐ Number of Fragments = 2 ; Frag#1: Offset=0, MF=1 ; Frag#2: Offset=1500, MF=0
- ☒ Number of Fragments = 3 ; Frag#1: Offset=0, MF=1 ; Frag#2: Offset=185, MF=1; Frag#3: Offset=370, MF=0
- ☐ Number of Fragments = 2 ; Frag#1: Offset=0, MF=0 ; Frag#2: Offset=1500, MF=1
- ☐ None of the above

 **Will be reviewed**

34

Question

- ☐ a)
- ☐ b)
- ☒ c)
- ☐ d)
- ☐ e)

 **Will be reviewed**

35

Consider TCP receiver waits for the segment with sequence number 1000. The sender sends two segments back-to-back with sequence numbers: 1000 contains 500 bytes, 1500 contains 500 bytes. If the first segment is lost and the second arrives, the receiver sends:

- ☐ ACK 1000
- ☒ ACK 1500
- ☐ ACK 2000
- ☐ ACK 2500
- ☐ None of the above

 **Will be reviewed**

36

Which of the following is NOT true about RIP?

- ☐ RIP counts the number of subnets traversed along the shortest path from source router to destination subnet including the destination subnet.
- ☐ The maximum cost of a path is limited to 15.
- ☐ Routing updates are exchanged between neighbors approximately every 30 seconds using RIP response message.
- ☒ RIP requires each node to first obtain a complete map of the network before running the algorithm.
- ☐ None of the above

 **Will be reviewed**

37

Question

- ☐ Yes
- ☒ No

 **Will be reviewed**

38

A mail server runs an SMTP service on port 25. At a given time, the server has three active SMTP connections, all destined for port 25, established by three different clients. Which of the following is used to deliver incoming segments to the correct connection?

- ☐ Matching only the destination IP address and destination port number
- ☐ Matching only the source IP address and source port number
- ☐ Matching only the source IP address and destination port number
- ☐ Matching only the source port number and destination IP address
- ☒ Matching the source IP address, source port number, destination IP address, and destination port number

☐ None of the above

 **Will be reviewed**

39

Which of the following is correct ?

- ☐ IMAP provides commands to allow users to create folders and move messages from one folder to another.
- ☐ IMAP provides commands that allow users to search remote folders for messages matching specific criteria.
- ☐ Unlike POP3, an IMAP server maintains state information across IMAP sessions.
- ☐ IMAP has commands that permit a user agent to obtain components of messages.
- ☒ All of the above
- ☐ None of the above

 **Will be reviewed**

40

Question

- ☒ Yes
- ☐ No

 **Will be reviewed**

41

Three nodes X, Y, and Z share a common broadcast channel and use CSMA/CD. Node X starts transmitting at 0 ms, Nodes Y and Z attempt to transmit at 2 ms and 3 ms, respectively. The propagation delays between X–Y, Y–Z, and X–Z are 1 ms, 1 ms, and

2 ms, respectively. Assume the packet transmission time is 4 ms. Which statement is correct?

- ☐ Node Y postpones transmission, node Z postpones transmission, and no collision occurs.
- ☒ Node Y transmits at 2 ms, node Z transmits at 3 ms, and a collision occurs.
- ☐ Node Y postpones transmission, node Z transmits at 3 ms, and no collision occurs.
- ☐ Node Y transmits at 2 ms, node Z postpones transmission, and a collision occurs.
- ☐ None of the above

 **Will be reviewed**

42

Which of the following is NOT a challenge of P2P application architecture?

- ☐ Security due to the distributed and open nature
- ☐ Incentivizing users to share bandwidth, storage, and computation
- ☒ ISP friendliness and asymmetrical bandwidth usage
- ☐ Infrastructure-intensive requirements such as data centers
- ☐ All of the above
- ☐ None of the above

 **Will be reviewed**

43

Which of the following statements correctly describes ISP network structures?

- ☐ Network Structure#1 consists of multiple global ISPs, and Network Structure#3 extends the hierarchy by adding IXPs networks.

- Network Structure#1 is based on dial-up access, Network Structure#2 is based on DSL access,
- ☐ Network Structure#3 uses cable networks, Network Structure#4 uses FTTH, and Network Structure#5 is entirely wireless.
- Network Structure#2 introduces multiple global transit ISPs, Network Structure#3 introduces multiple
- ☒ competing regional ISPs, and Network Structure#5 allows large content provider networks to connect directly to access ISPs.
- ☐ Network Structure#3 adds peering links through Internet Exchange Points, and Network Structure#4 allows content provider networks to connect directly to access ISPs.
- ☐ None of the above

 **Will be reviewed**

44

Which of the following is a characteristic of P2P file distribution?

- ☐ Distribution time increases linearly with the number of peers
- ☐ The server must send a complete copy of the file to each peer.
- ☒ Distribution is self-scalable due to peers acting as redistributors.
- ☐ More secure than client-server distribution.
- ☐ None of the above

 **Will be reviewed**

45

Which statement correctly compares HFC, FTTH, DSL, and Dial-Up?

- ☐ DSL and Dial-Up are shared, while HFC and FTTH are dedicated
- ☐ HFC and Dial-Up are dedicated, while DSL and FTTH are shared
- ☒ HFC is shared, DSL is dedicated, Point-to-Point (Direct) FTTH is dedicated, while Dial-Up is dedicated and has the lowest rate

- ☐ DSL and HFC are shared, while Dial-up and FTTH are dedicated
- ☐ None of the above

 **Will be reviewed**

46

Consider router D, which is a neighbor of routers A and B. Router D has an existing path to subnet Z via router B with a hop count of 7 from D to Z. Later, router D receives a routing advertisement from router A indicating that subnet Z is 6 hops away from router A. In this case, classic RIP, which keeps the existing path unless the new one is better, will:

- ☐ Keep the existing path.
- ☐ Update the routing table to route via router A instead of router B.
- ☒ Cause a count-to-infinity problem.
- ☐ Cause an oscillation problem.
- ☐ None of the above

 **Will be reviewed**

47

Which of the following is correct about IP and MAC addresses ?

- ☐ MAC address allows communication across different subnets, while IP address is used within a LAN.
- ☒ IP address is used to identify interfaces on the same LAN, while MAC address is used globally.
- ☐ Unlike IP address, MAC address enables device to receive frames at the link layer without the host being interrupted by irrelevant frames.
- ☐ IP address has flat structure while MAC address has hierarchical structure .
- ☐ All of the above

☐ None of the above

 **Will be reviewed**

48

IPv6/IPv4 Node A wants to send data to IPv6/IPv4 Node B through an IPv4-only router. Which statement is TRUE for Tunneling Scenario?

- ☐ Node A drops the datagram.
- ☐ Node B must downgrade to IPv4.
- ☐ Node A automatically converts IPv6 headers to IPv4 headers.
- ☒ Node A encapsulates the IPv6 datagram inside an IPv4 datagram.
- ☐ None of the above



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